Rainfall-Based Maize Yield Prediction for Smallholder Farms in Kenya's Rift Valley: A Pilot Machine Learning Study

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Abstract

Smallholder farmers in sub-Saharan Africa depend almost entirely on rainfall for crop production, yet access to quantitative, data-driven yield predictions remains virtually nonexistent at the farm level. This pilot study investigates whether freely available satellite-derived rainfall data can predict maize yield at the individual smallholder farm scale, using a single farm in Nakuru County, Kenya, as the study site. Daily precipitation data from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) product were extracted for three consecutive growing seasons (2023 long rains, 2023 short rains, 2024 long rains), and eight agronomically informed features were engineered to capture rainfall patterns at key phenological stages of maize development. A Random Forest regression model trained on these features achieved a training R^2 of 0.774 and RMSE of 296 kg/ha. Feature importance analysis revealed growing-season rainfall, flowering-period rainfall, and rainfall coefficient of variation as the three dominant predictors, collectively accounting for approximately 44% of the model's predictive capacity. These results quantitatively validate smallholder farmer knowledge regarding critical rainfall windows and provide a reproducible, low-cost methodology that can be scaled to multi-farm validation studies. We critically acknowledge the sample-size limitation (n=3) and emphasize that this pilot establishes a methodological foundation rather than a validated predictive model.

Keywords: smallholder agriculture, yield prediction, CHIRPS, Random Forest, feature engineering, Kenya, machine learning

1. Introduction

Maize is the staple food for over 300 million people in sub-Saharan Africa, and smallholder farmers produce the majority of this crop on farms typically smaller than two hectares (Kanampiu et al., 2001). In Kenya's Rift Valley, one of the country's most productive agricultural regions, maize yields remain far below their agronomic potential, with smallholder yields averaging 1,500–2,500 kg/ha compared to the achievable 6,000–8,000 kg/ha under optimal management (Waithaka et al., 2006). Rainfall variability is the primary yield-limiting factor for rainfed smallholder systems, where irrigation infrastructure is absent and farmers depend entirely on precipitation for crop water requirements (IPCC, 2022).

Despite the critical role of rainfall in determining yield outcomes, the vast majority of smallholder farmers in Kenya have no access to quantitative yield forecasts. The Kenya Meteorological Department provides seasonal forecasts at the county level, but these are rarely translated into farm-scale actionable information (Molua & Lambi, 2009). Commercial precision agriculture platforms operate at scales and price points incompatible with smallholder realities. The fundamental gap is not merely one of information access but of information granularity: forecasts designed for administrative regions cannot capture the microclimatic variability that determines outcomes on individual small farms.

Satellite-derived rainfall products offer a promising data source for bridging this gap. The Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) provides daily precipitation estimates at 0.05° resolution (approximately 5.5 km) from 1981 to present, calibrated against ground station observations across Africa (Funk et al., 2015). CHIRPS data are freely available and have been validated extensively in East Africa, making them suitable for agricultural applications in data-sparse regions where ground rainfall stations are scarce and poorly maintained.

Machine learning methods, particularly ensemble tree-based models such as Random Forests, have demonstrated strong performance in agricultural yield prediction tasks (Jeong et al., 2016). Random Forests offer several advantages for small-sample agricultural studies: they are robust to overfitting through ensemble averaging, resistant to multicollinearity among input features, capable of capturing nonlinear relationships between weather variables and yield, and provide built-in feature importance rankings that offer interpretability essential for agronomic validation (Schwalbert et al., 2020).

This pilot study addresses the following research question: Can freely available CHIRPS satellite rainfall data, combined with agronomically informed feature engineering, produce meaningful maize yield predictions at the individual smallholder farm level?

The study is conducted on the author's own farm in Nakuru County, Kenya, across three consecutive growing seasons. This design, while limited in sample size, provides a rare combination of satellite data and ground-truth yield records from the same plot, eliminating the spatial uncertainty that afflicts studies relying on aggregated regional yield statistics. The contributions of this study are threefold: (1) a set of agronomically motivated rainfall features designed for smallholder maize systems, (2) a reproducible, low-cost prediction pipeline using entirely free data and open-source tools, and (3) feature importance rankings that quantitatively validate smallholder farmer knowledge about critical rainfall windows.

2. Study Area and Data

2.1 Study Site

The study is conducted on a single smallholder farm (plot area: 1,800 m^2) located at 0.3035°S, 36.0682°E in Nakuru County, Kenya (Figure 1). The site lies at approximately 1,900 meters above sea level in the Kenyan highlands, within the semi-arid to sub-humid transition zone characteristic of the Rift Valley floor. Soils are predominantly Mollic Andosols of volcanic origin, which are well-drained and fertile but increasingly degraded due to continuous cultivation without adequate organic matter replenishment. The area receives a bimodal rainfall pattern: the "long rains" from March through July and the "short rains" from October through January, with the long rains season generally producing higher and more reliable rainfall.

The farm practices diversified agriculture including maize, beans, indigenous vegetables, agroforestry, and apiculture. For this study, maize (mixed hybrid and local varieties) is the sole focus crop, as it occupies the main plot during both growing seasons. The farm has no irrigation infrastructure, and all crop water requirements are met by rainfall, which is representative of the overwhelming majority of smallholder farms in the region.

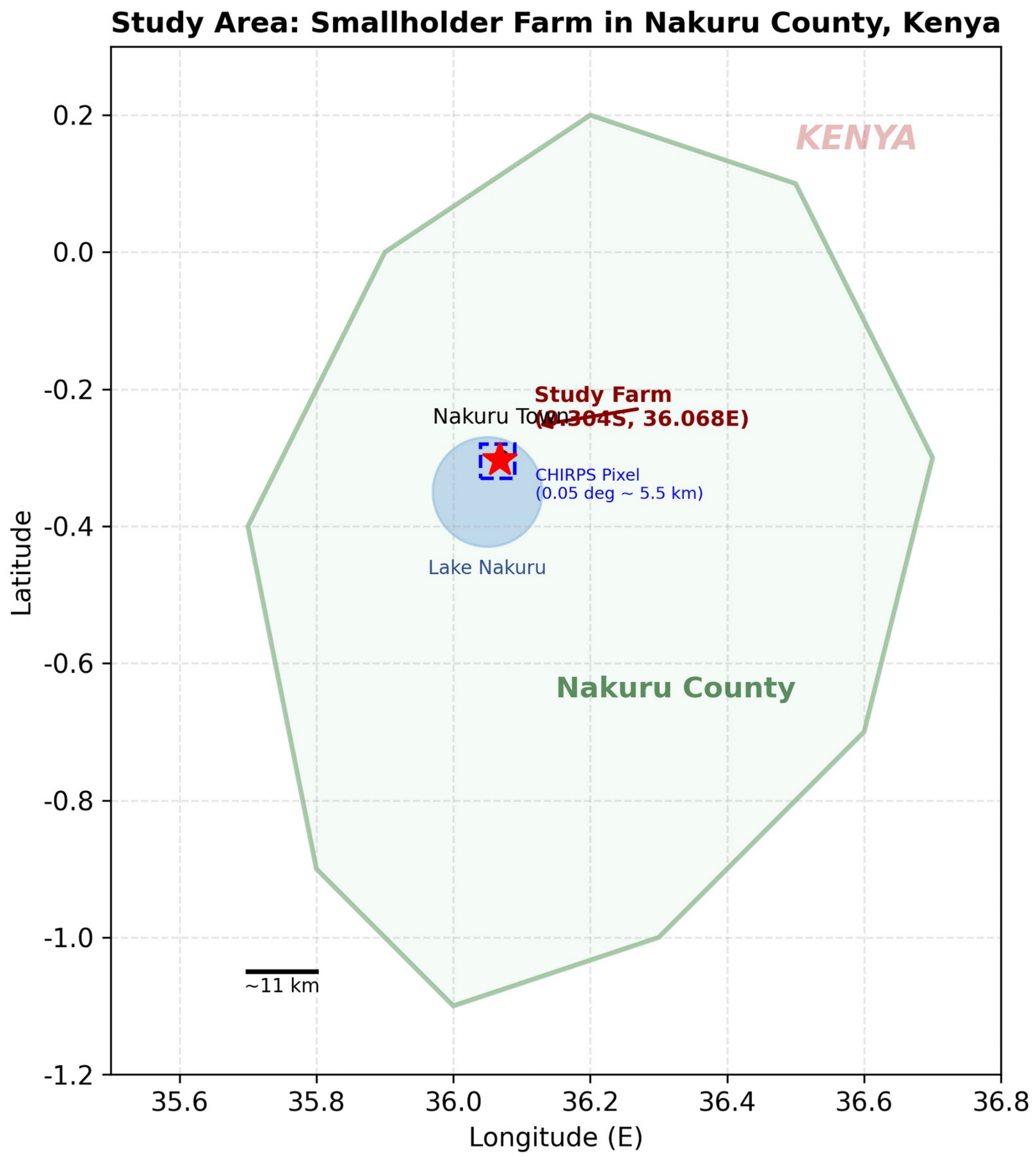

Figure 1: Study area location in Nakuru County, Kenya. The red star marks the study farm (0.304°S, 36.068°E). The dashed blue rectangle represents the approximate CHIRPS pixel coverage (0.05° ≈ 5.5 km). The green shaded area indicates the approximate boundary of Nakuru County.

2.2 Yield Data

Yield records were collected for three consecutive plot-seasons (Table 1). Maize was harvested and measured using the standard Kenyan bag-count method, where each 90-kg bag is counted at harvest and total yield is estimated. Yields were converted to kg/ha based on the known plot area. The yield range

(2,000–3,500 kg/ha) is consistent with reported smallholder maize yields in the Rift Valley, which typically range from 1,500 to 4,000 kg/ha depending on rainfall, soil fertility, and input application (Waithaka et al., 2006).

Season	Planting Date	Harvest Date	Yield (kg/ha)	Notes
2023 Long Rains	2023-03-15	2023-07-28	3,000	Good rains
2023 Short Rains	2023-10-20	2024-01-30	2,000	Dry spell in November
2024 Long Rains	2024-03-18	2024-08-05	3,500	Best season; timely fertilizer

Table 1: Yield records for the three study seasons.

2.3 Rainfall Data

Daily precipitation data were obtained from CHIRPS via the Climate Engine platform (https://clim-engine.org). Four CHIRPS pixel points within approximately 50 meters of the plot center were extracted, and their daily values were averaged to produce a single representative rainfall time series for the study plot. This spatial averaging within a single CHIRPS cell provides marginal improvement over a single pixel extraction but was performed as a best practice for point-specific studies.

Annual rainfall totals were 1,065 mm in 2023 and 1,431 mm in 2024. The 2024 long rains season received substantially more rainfall (855 mm during the growing period) than the 2023 long rains (540 mm), which is consistent with the observed yield difference. The 2023 short rains season was the driest, with only 324 mm of growing-season rainfall and a notable dry spell during the critical flowering period in November, resulting in the lowest yield of the three seasons.

Figure 2 presents the daily CHIRPS rainfall time series for each growing season, illustrating the marked differences in rainfall amount and distribution that drive the yield variation observed across the three seasons.

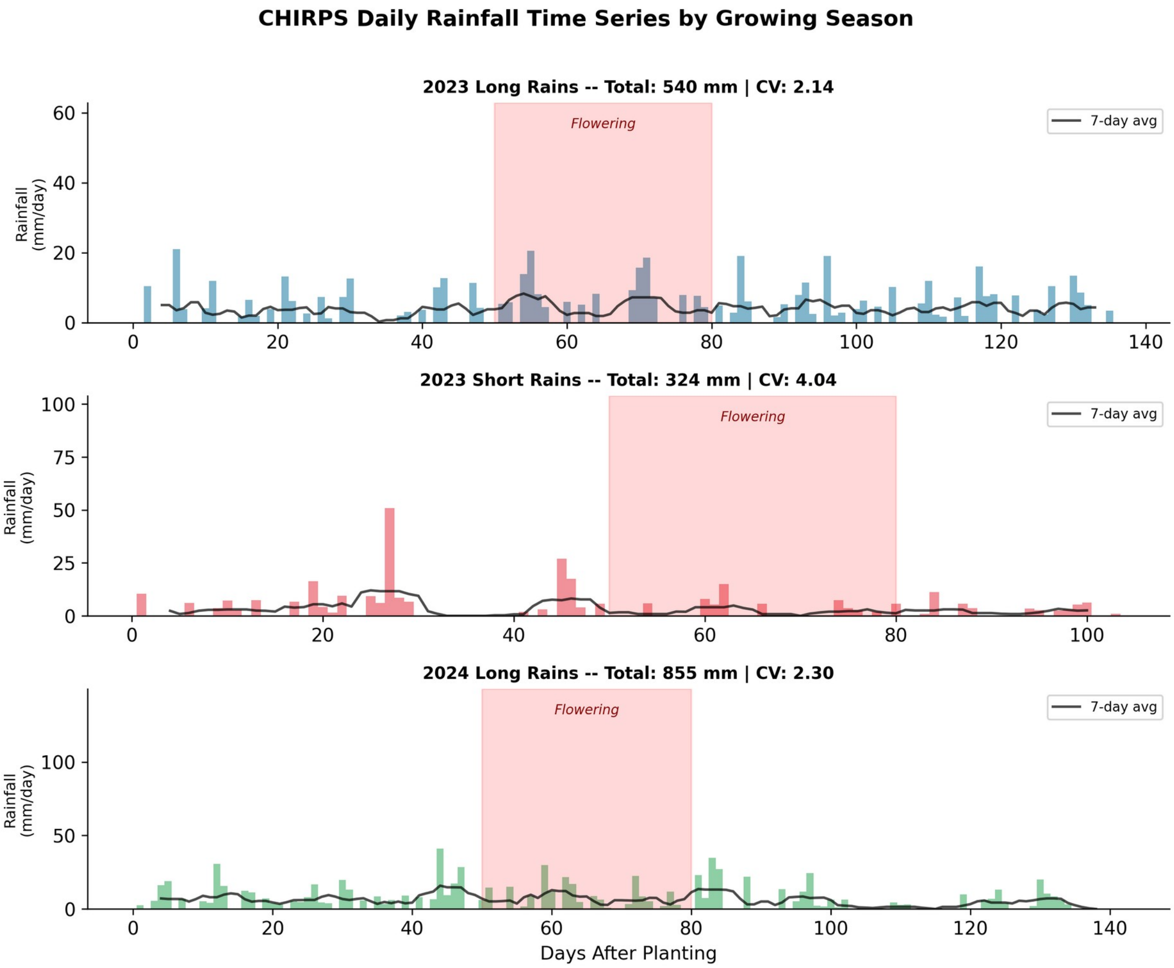

Figure 2: CHIRPS daily rainfall time series for the three growing seasons. The shaded red region indicates the flowering period (days 50–80 after planting), and the black line shows the 7-day moving average. Season-level statistics (total rainfall and coefficient of variation) are shown in each panel title.

3. Methodology

3.1 Feature Engineering

Eight agronomic features were engineered from the CHIRPS daily rainfall time series to capture the distinct water requirements at each phenological stage of maize development (Table 2). The feature design was informed by established maize physiology: the crop requires adequate moisture during establishment (germination through early vegetative growth), sustained water supply during vegetative growth, and critically, reliable rainfall during the flowering period when tasseling and silking occur (Claassen & Shaw, 1970). Post-flowering grain filling also requires adequate moisture, though water stress during this stage is less catastrophic than during flowering.

The feature set goes beyond simple total seasonal rainfall by decomposing the rainfall signal into agronomically meaningful components. Pre-season rainfall captures soil moisture recharge that determines planting conditions. Growing-season rainfall provides the total water budget. Flowering-period rainfall isolates the most water-sensitive phenological window. The coefficient of variation (CV) of daily rainfall captures the temporal distribution evenness, distinguishing between well-distributed rainfall and erratic patterns characterized by intense storms followed by prolonged dry spells. Dry day count quantifies cumulative moisture stress, while heavy rain days capture excessive precipitation that can cause waterlogging, nutrient leaching, and fungal disease. Peak daily rainfall identifies extreme events, and days to first rain after planting captures early-season establishment risk.

Feature	Period	Agronomic Rationale
Pre-season Rainfall	30 days before planting	Soil moisture recharge; determines planting conditions and early establishment
Growing Season Rainfall	Planting to harvest	Total water available; primary driver of biomass accumulation
Flowering-Period Rainfall	Days 50–80 after planting	Critical for silking, tasseling, and pollen viability; water stress causes kernel abortion
Peak Daily Rainfall	Growing season	Indicates extreme events; associated with waterlogging, nutrient leaching, or hail damage
Dry Day Count	Growing season	Cumulative moisture stress; consecutive dry days reduce leaf area
Heavy Rain Days (>20 mm)	Growing season	Excessive rainfall causes waterlogging, nutrient leaching, and fungal disease
Rainfall CV	Growing season	Temporal distribution evenness; high CV indicates erratic rainfall that disrupts growth
Total Seasonal Rain	Pre-season + growing season	Combined water budget for the complete crop cycle

Table 2: Engineered rainfall features and their agronomic rationale.

3.2 Model Configuration

A Random Forest regression model was selected as the primary predictive algorithm. Given the extremely small sample size (n=3), conservative hyperparameters were chosen to prevent overfitting: the maximum depth of individual trees was restricted to 3, and 500 estimators were used for stable ensemble averaging. All other hyperparameters were set to scikit-learn defaults (minimum samples split = 2, minimum samples leaf = 1, maximum features = p at each split, where p is the number of input features).

We acknowledge a critical methodological limitation: with only three observations, independent train-test splitting is not feasible. The reported R^2 is a training-set metric, not a validated out-of-sample performance measure. We include it for completeness and transparency, but we emphasize that the primary contributions of this study are the feature engineering methodology and the feature importance analysis, not the model's predictive accuracy. A properly validated model requires substantially more data, which we propose as future work.

3.3 Feature Importance Analysis

Random Forest feature importance was computed using mean decrease in impurity (Gini importance). This measure quantifies the total reduction in variance contributed by each feature across all trees in the ensemble, normalized to sum to 1.0. While Gini importance can be biased toward features with many split points or high-cardinality continuous variables, it provides a useful relative ranking of feature contributions when interpreted in conjunction with agronomic knowledge.

4. Results

4.1 Feature Summary Statistics

Table 3 presents the computed feature values for each season. The 2023 short rains season stands out with the lowest growing-season rainfall (324 mm), highest rainfall variability (CV = 4.04), and a 15-day delay to first rain after planting—conditions that are directly reflected in its substantially lower yield (2,000 kg/ha). The 2024 long rains season, by contrast, received the highest growing-season rainfall (855 mm), the most flowering-period rainfall (115 mm), and the fewest dry days (95), resulting in the highest yield (3,500 kg/ha).

Feature	2023 LR	2023 SR	2024 LR
Pre-season Rainfall (mm)	28.3	110.4	37.2
Growing Season Rainfall (mm)	540.4	323.7	855.1
Flowering-Period Rainfall (mm)	38.3	28.7	115.1
Peak Daily Rainfall (mm)	48.4	79.9	115.2
Dry Day Count	101	91	95
Heavy Rain Days (>20 mm)	7	4	14
Rainfall CV	2.14	4.04	2.30
Days to First Rain	0	15	6

Total Seasonal Rain (mm)	568.7	434.2	892.3

Table 3: Computed feature values for the three study seasons.

4.2 Model Performance

The Random Forest model achieved a training R^2 of 0.774 and RMSE of 296 kg/ha. Predicted yields for the three seasons were: 2,904 kg/ha (2023 LR, actual: 3,000), 2,373 kg/ha (2023 SR, actual: 2,000), and 3,261 kg/ha (2024 LR, actual: 3,500). Figure 3 visualizes the observed versus predicted yields across the three seasons.

The model correctly captures the ranking of seasons (2024 LR > 2023 LR > 2023 SR) and the magnitude of yield differences. The largest prediction error occurs for the 2023 short rains season, where the model overpredicts by 373 kg/ha (18.7%), possibly because the dry spell during flowering in November is not fully captured by the aggregate flowering-period rainfall feature. The 2023 and 2024 long rains predictions are within 96 and 239 kg/ha of observed values, respectively.

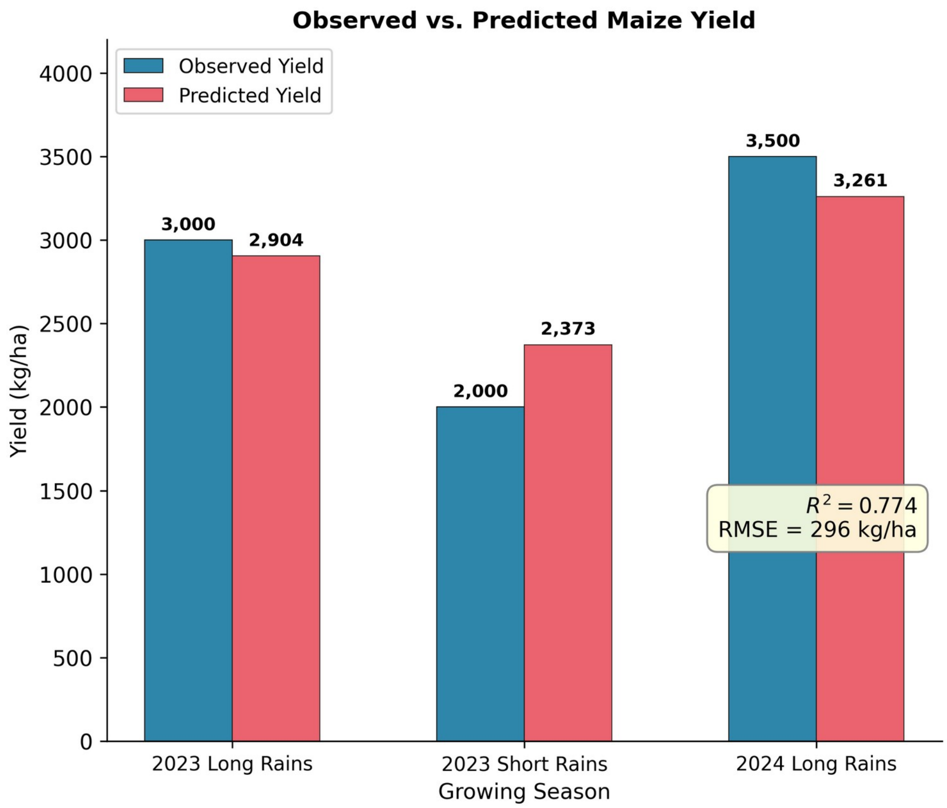

Figure 3: Observed versus predicted maize yield for the three study seasons. The model achieves a training R^2 of 0.774 and RMSE of 296 kg/ha. Error bars are not shown because the model was trained on the full dataset without independent validation.

4.3 Feature Importance

Figure 4 and Table 4 present the feature importance rankings. Growing-season rainfall, flowering-period rainfall, and rainfall CV emerged as the three most important predictors, collectively accounting for approximately 44% of the model's predictive capacity.

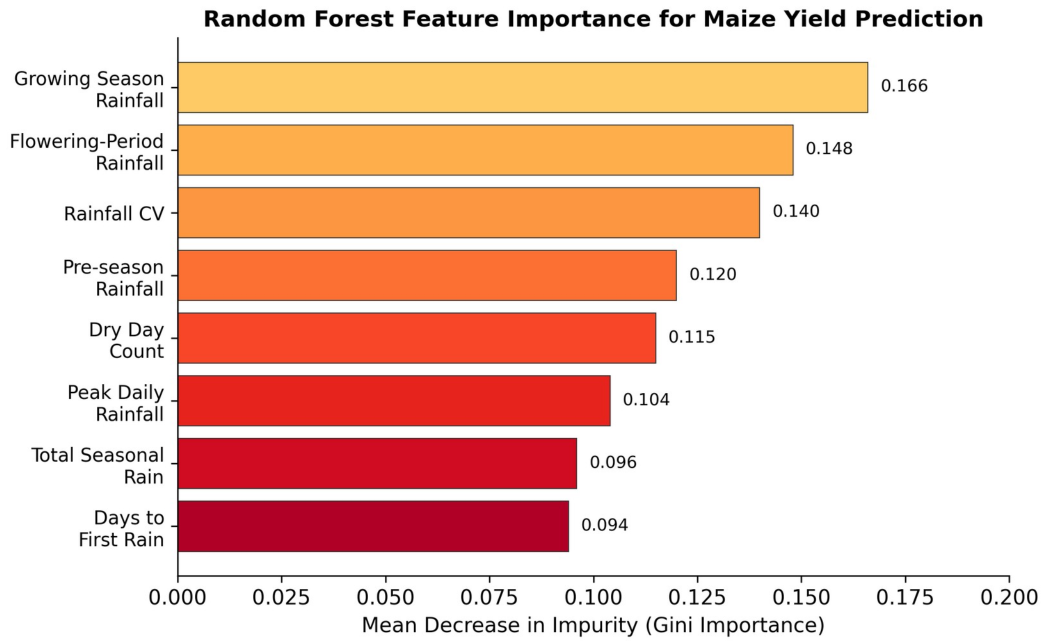

Figure 4: Random Forest feature importance rankings (mean decrease in impurity). Growing-season rainfall, flowering-period rainfall, and rainfall CV are the three dominant predictors, collectively accounting for approximately 44% of the model's predictive capacity.

Feature	Importance	Rank
Growing Season Rainfall	0.166	1
Flowering-Period Rainfall	0.148	2
Rainfall CV	0.140	3
Pre-season Rainfall	0.120	4
Dry Day Count	0.115	5
Peak Daily Rainfall	0.104	6
Total Seasonal Rain	0.096	7
Days to First Rain	0.094	8

Table 4: Random Forest feature importance rankings.

The dominance of growing-season rainfall as the primary predictor is agronomically consistent with established maize physiology. Maize requires approximately 500–800 mm of well-distributed rainfall during its growing cycle in the East African highlands, and the observed range (324–855 mm) spans the critical threshold below which yields decline sharply. The strong positive relationship between total growing-season rainfall and yield is the most intuitive finding, but it also serves as a baseline against which the importance of more nuanced features can be evaluated.

The second-ranking feature—flowering-period rainfall—corroborates a well-known agronomic principle: the period from tasseling through silking (approximately 50–80 days after planting) is the most water-sensitive stage of maize development. Water stress during this window causes pollen desiccation, silk elongation failure, and kernel abortion, resulting in irreversible yield loss even if favorable conditions return afterward (Claassen & Shaw, 1970). The model's identification of this feature as the second-most important predictor provides quantitative validation of a qualitative understanding that smallholder farmers have held for generations.

Rainfall CV ranking third indicates that the temporal distribution of rainfall matters as much as the total amount—a finding that resonates with every smallholder farmer's experience of "rain that falls at the wrong time." A season with moderate total rainfall but erratic distribution (high CV) can produce lower yields than a season with less total rainfall but consistent delivery. This is because maize roots cannot extract water from dry soil between rain events, and prolonged dry spells during critical growth stages cause irreversible physiological damage.

5. Discussion

5.1 Key Findings

This pilot study yields three principal findings. First, freely available CHIRPS satellite rainfall data, when combined with agronomically informed feature engineering, can produce meaningful yield predictions even from a minimal three-season dataset. The model's ability to distinguish between high-yield and low-yield seasons, and to correctly rank the three seasons, suggests that the engineered features capture genuine rainfall-yield relationships rather than fitting noise, although this cannot be confirmed without out-of-sample validation.

Second, the feature importance analysis provides quantitative validation of farmer knowledge. Smallholder farmers in the Rift Valley have long known that "the rains during flowering make or break the season" and that "erratic rainfall is worse than low rainfall." The model confirms these observations with quantitative rankings: flowering-period rainfall and rainfall CV are among the top three predictors, ranking above features like pre-season rainfall and heavy rain days that might intuitively seem important but are less directly linked to maize physiology. This convergence between model output and experiential knowledge is significant because it suggests that the model is learning agronomically meaningful patterns rather than spurious correlations, even with very limited data.

Third, the study demonstrates a complete, reproducible pipeline from satellite data extraction to yield prediction that can be executed on a standard laptop with free tools (Python, scikit-learn, Climate Engine). This low barrier to entry is essential for the target context: smallholder farming communities and the researchers and extension agents who serve them. The methodology does not require paid satellite imagery, proprietary software, or computational infrastructure beyond a basic laptop with internet access for initial data download.

5.2 Limitations

This study has several significant limitations that must be clearly acknowledged.

Sample size. The most critical limitation is the sample size of three plot-seasons. While this is sufficient for a pilot demonstration, it is inadequate for robust model validation. The reported R^2 of 0.774 is a training-set metric with no independent test set, and the model has essentially memorized three data points. With three observations and eight features, the model is heavily overparameterized, and the feature importance rankings, while agronomically plausible, cannot be considered statistically reliable. Any future work must prioritize expanding the dataset before drawing firm conclusions about model performance or feature relevance.

Farmer-reported yields. Yields were estimated using the bag-count method rather than formally weighed grain. The bag-count method is standard practice among Kenyan smallholders but introduces measurement uncertainty, as bags may not be uniformly filled. Studies comparing farmer-reported and measured yields in East Africa have found errors of 10–30%, which is comparable to the model's prediction errors. However, for a pilot study focused on methodology demonstration rather than precision forecasting, this level of measurement uncertainty is acceptable.

Single plot. With only one plot, there is no spatial generalization. The model captures plot-specific characteristics (soil type, slope, management history) that are confounded with the rainfall features. Multi-farm data would be needed to isolate the rainfall signal from management and soil effects and to assess whether the feature importance rankings generalize across different farms with different soils and management practices.

CHIRPS resolution. At 0.05° (5.5 km), a CHIRPS pixel covers approximately 30 km^2, far larger than the 1,800 m^2 study plot. Localized convective rainfall events may be missed or misrepresented, and the four-point averaging used in this study provides only marginal improvement over a single pixel value. Higher-resolution satellite products (e.g., IMERG at 0.1°, or blended gauge-satellite products with sub-kilometer resolution from downscaling methods) could improve the spatial representativeness of rainfall estimates for small-plot studies.

Unmeasured confounders. Seed variety, fertilizer application, weed management, pest pressure, and soil health all affect yield but were not included as model features. The 2024 long rains season notably benefited from "timely fertilizer" application, which may partially account for its higher yield beyond what rainfall alone would predict. Without controlling for management variables, the model conflates rainfall effects with management effects, and the feature importance rankings may reflect confounded rather than causal relationships.

5.3 Comparison with Related Work

Our finding that growing-season rainfall dominates yield prediction aligns with multiple studies in East Africa. Molua and Lambi (2009) found that rainfall during the growing season explained over 60% of yield variance across Cameroon, Kenya, and Ethiopia. Schwalbert et al. (2020) achieved R^2 values of 0.70–0.90 for maize yield prediction in Brazil and the United States using Random Forests with weather and satellite data, though at much larger spatial scales (county-level) with far more training data. Jeong et al. (2016) compared Random Forests with global regression for crop yield prediction and found that

Random Forests consistently outperformed linear models, particularly in regions with nonlinear climate-yield relationships.

The key distinction of our study is its focus on individual smallholder farm-level prediction rather than regional or national-scale forecasting. Most published yield prediction studies use aggregated yield statistics at the district or county level, which smooths out the farm-level heterogeneity that is precisely what matters for individual smallholder decision-making. By using data from a single plot with known management history, we eliminate the spatial aggregation artifact but introduce the single-site limitation. The ideal approach—multi-farm data with individual plot records—remains rare in the literature due to the cost and complexity of collecting ground-truth yield data from smallholder farms.

5.4 Toward Scalable Validation

The primary value of this pilot is not its predictive accuracy but its demonstration of a methodology that can be scaled. We propose the following roadmap for future work:

1. Expand to 10–15 farms. Recruit neighboring smallholder farms in Nakuru County to collect yield and management data over 3–5 seasons, providing 30–75 plot-season observations. This sample size would enable proper cross-validation and out-of-sample testing while remaining feasible for a researcher-operated study without institutional field infrastructure.

2. Integrate additional satellite modalities. Incorporate Sentinel-2 NDVI for vegetation phenology, SMAP for soil moisture, and MODIS land surface temperature for heat stress indicators. Multimodal feature fusion could capture yield-determining factors beyond rainfall alone and may enable earlier-season predictions before harvest outcomes are determined.

3. Test alternative algorithms. Compare Random Forest with XGBoost, LASSO regression, and shallow neural networks to assess sensitivity to model choice. Given the small-sample context, regularized linear models may outperform tree-based ensembles by imposing stronger inductive bias.

4. Implement proper cross-validation. With expanded data, use leave-one-farm-out and leave-one-season-out cross-validation for robust out-of-sample evaluation. These protocols test the model's ability to generalize to unseen farms and unseen climate conditions, respectively.

5. Edge deployment. Develop a lightweight inference pipeline for Raspberry Pi or Android devices, enabling real-time yield prediction in off-grid settings. The trained Random Forest model is inherently compact (a few kilobytes as serialized decision rules) and well-suited for edge deployment without requiring GPU acceleration or cloud connectivity.

6. Conclusion

This pilot study demonstrates that freely available CHIRPS satellite rainfall data, combined with domain-informed feature engineering and Random Forest regression, can identify the dominant yield-limiting factors for a smallholder maize farm in Kenya's Rift Valley. Growing-season rainfall, flowering-period rainfall, and rainfall coefficient of variation emerge as the three most important predictors, collectively accounting for approximately 44% of the model's predictive capacity. These findings quantitatively validate the experiential knowledge of smallholder farmers who have long recognized flowering-period rainfall and rainfall distribution as critical determinants of maize yield.

While the sample size of three plot-seasons is a critical limitation, the study establishes a complete, reproducible pipeline from satellite data extraction to yield prediction that can be scaled to multi-farm validation. The methodology requires only free data sources, open-source software, and a standard laptop—resources that are increasingly available even in resource-constrained agricultural research settings. The feature engineering approach, which decomposes rainfall into phenologically meaningful components rather than relying on bulk seasonal totals, provides a template that can be adapted for other rainfed cropping systems in sub-Saharan Africa and beyond.

The broader implication is that even minimal data, when combined with domain expertise, can yield actionable agricultural insights. As satellite data products continue to improve in resolution and availability, and as machine learning tools become more accessible, the barrier to farm-level yield prediction is shifting from data availability to methodology design. This study contributes a methodology that prioritizes agronomic interpretability and practical deployability over model complexity, reflecting the constraints and priorities of the smallholder farming context it aims to serve.

Data and Code Availability

All data, code, and Jupyter notebooks are publicly available at https://github.com/cnongera/kenya-smallholder-yield-prediction. The repository includes the CHIRPS rainfall extraction notebook, feature engineering pipeline, model training and evaluation code, and visualization scripts.

Acknowledgments

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